Introduction to Probability and Statistics

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Statistics for Engineers



Course Code: STAT 251
Course Title: Introduction to Probability and Statistics
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Name of the Topics of this Course

- Statistics and Data Description
- Probability
- Probability Distribution
- Stochastic Process and Queueing Theory



Probability and Set Theory

Identity Law	:	$A \cup \varnothing = A,\ A \cup U = U$
		$A \cap U = A,\ A \cap \varnothing = \varnothing$
De Mograns Law	:	$(A \cup B)' = A' \cap B'$
		$(A \cap B)' = A' \cup B'$
Associative Law	:	$A \cup (B \cup C) = (A \cup B) \cup C$
		$A \cap (B \cap C) = (A \cap B) \cap C$
Distributive Law	:	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
		$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$



Probability

Conditional Probability: $\frac{n(A \cap B)}{n(B)} = \frac{\frac{n(A \cap B)}{n}}{\frac{n(B)}{n}} = \frac{P(A \cap B)}{P(B)} = P(A|B)$

$P(A \cap B) = P(A)P(B)$ if the event A and B are independent. Also, $P(A|B) = P(A)$ and $P(B|A) = P(B)$.

$P(A'|B) = 1 - P(A|B)$



Probability

Q. What is meant by conditional probability and Tree diagram? State and prove the Bayes theorem. For any events A and B with $P(A) > 0$, and $P(B|A) > P(B)$ then show that, $P(B'|A) < P(B')$.



Probability

Example A certain type of refrigerator comes in three sizes, each of which is available with or without an ice-maker. The accompanying table gives the relevant probabilities for a randomly selected purchaser:

		Size		
		17.5-ft^3	21-ft^3	24.5-ft^3
Ice-maker	Yes	0.12	0.17	0.24
	No	0.18	0.13	0.16

a) What is the probability that the purchaser selected a $21 - ft^3$ refrigerator? A refrigerator with an ice-maker?

b) Given that the purchaser select a model with an ice-maker, what is the probability it was a $21 - ft^3$ model? How does this compare to the unconditional probability of selecting a $21 - ft^3$ model?

c) Given that the purchaser selected a model with at least $20 - ft^3$ of space, what is the probability that the selected refrigerator did not have an ice-maker?



Probability

Solution: Here,

		Size			
		17.5-ft^3	21-ft^3	24.5-ft^3	Total
Ice-maker	Yes	0.12	0.17	0.24	0.53
	No	0.18	0.13	0.16	0.47
	Total	0.30	0.30	0.40	1.00

(a) $P(21 - ft^3\text{Refri.}) = 0.30$, and $P(\text{Refri. with Ice-maker}) = 0.53$

(b) $P(21 - ft^3\text{Refri.}|\text{with Ice-maker}) = \dfrac{P(21 - ft^3\text{Refri.} \cap \text{Refri. with Ice-maker})}{P(\text{Refri. with Ice-maker})}$

$$P(21 - ft^3\text{Refri.}|\text{with Ice-maker}) = \frac{0.17}{0.53} = 0.321$$

Again, $P(21 - ft^3\text{Refri.}) = 0.30$

$P(21 - ft^3\text{Refri.}|\text{with Ice-maker}) > P(21 - ft^3\text{Refri.})$

(c) $P(\text{Refri. without Ice-maker}|\text{Refri. with at least } 20 - ft^3)$

$$= \frac{P(\text{Refri. without Ice-maker}) \cap P(\text{Refri. with at least} 20 - ft^3)}{P(\text{Refri. with at least 20-}ft^3)}$$

$$= \frac{0.13 + 0.16}{0.30 + .040} = \frac{0.29}{0.70} = 0.414$$



Bayes Theorem

The Law of Total Probability

$$P(B) = \sum_{i=1}^{k} P(A_i \cap B) = \sum_{i=1}^{k} P(B \cap A_i) = \sum_{i=1}^{k} P(B|A_i)P(A_i)$$

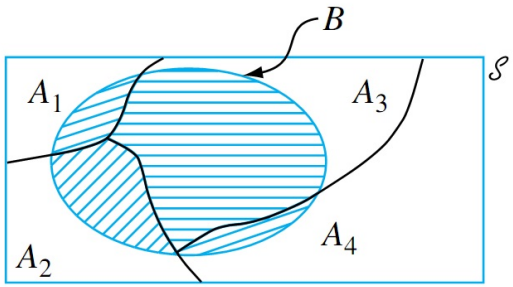


Figure: Partition of B by mutually exclusive and exhaustive A_i's.

Bayes Theorem Let $A_1, A_2, ..., A_k$ be a collection of k mutually exclusive and exhaustive events with prior probabilities $P(A_i)$ $(i = 1, ..., k)$. Then for any other event B for which $P(B) > 0$, the posterior probability of A_j given that B has occurred is: $P(A_j|B) = \dfrac{P(A_j \cap B)}{P(B)} = \dfrac{P(B|A_j)P(A_j)}{\sum_{i=1}^{k} P(B|A_i)P(A_i)}$

Probability

Example A company uses three different assembly lines—A_1, A_2, and A_3—to manufacture a particular component. Of those manufactured by line A_1, 5% need rework to remedy a defect, whereas 8% of A_2's components need rework and 10% of A_3's need rework. Suppose that 50% of all components are produced by line A_1, 30% are produced by line A_2, and 20% come from line A_3. If a randomly selected component needs rework, what is the probability that it came from line A_1? From line A_2? From line A_3?

Solution: Let B denote the event that a component needs rework. Then

$P(B|A_1) = 5\% = 0.05,\ P(B|A_2) = 8\% = 0.08,\ P(B|A_3) = 10\% = 0.10$

$P(A_1) = 50\% = 0.50,\ P(A_2) = 30\% = 0.30,\ P(A_3) = 20\% = 0.20$

$$P(B) = \sum_{i=1}^{3} P(B|A_i)P(A_i) = (0.05)(0.50) + (0.08)(0.30) + (0.10)(0.20) = 0.069$$

$$P(A_1|B) = \frac{P(B|A_1)P(A_1)}{P(B)} = \frac{(0.05)(0.50)}{0.069} = 0.362$$

$P(A_2|B) = 0.348$

$P(A_3|B) = 0.290$



Probability

Example A worker has asked her supervisor for a letter of recommendation for a new job. She estimates that there is an 80 percent chance that she will get the job if she receives a strong recommendation, a 40 percent chance if she receives a moderately good recommendation, a 10 percent chance if she receives a week recommendation. She further estimates that the probabilities that the recommendation will be strong, moderate, and week are 0.7, 0.2, and 0.1 respectively.

(a) How certain is she that she will receive the new job offer?

(b) Given that she does receive the offer, how likely should she feel that she received a strong recommendation; a moderate recommendation; a week recommendation?

(c) Given that she does not receive the job offer, how likely should she feel that she received a strong recommendation, a moderate recommendation; a week recommendation?



Solution Let,
S: denote the event that she will receive a strong recommendation,
M: denote the event that she will receive a moderately good recommendation,
W: denote the event that she will receive a week recommendation,
R: denote the event that she will receive the job offer.

$P(R|S) = *$, $P(R|M) = *$, $P(R|W) = *$, $P(R'|S) = 1 - P(R|S) = *$
$P(R'|M) = *$, $P(R'|W) = *$, $P(S) = *$, $P(M) = *$, $P(W) = *$

(a) $P(R) =?$
(b) $P(S|R) =?$, $P(M|R) =?$, $P(W|R) =?$
(c) $P(R') =?$, $P(S|R') =?$, $P(M|R') =?$, $P(W|R') =?$

